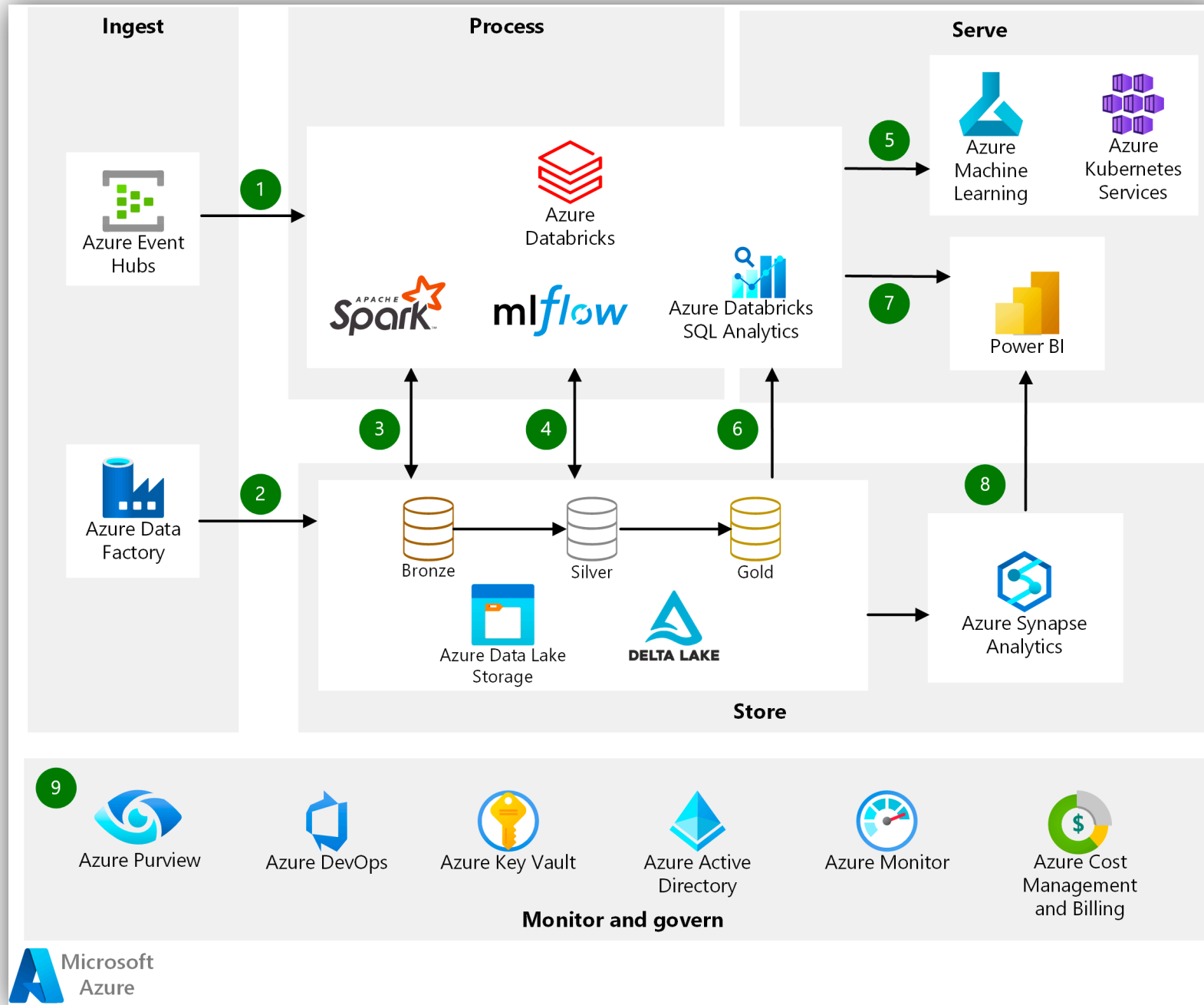


Azure Data Engineering

ADF + Databricks

Architecture

End to End Project Flow



What is Ingestion?

Data Files



Big Data Files



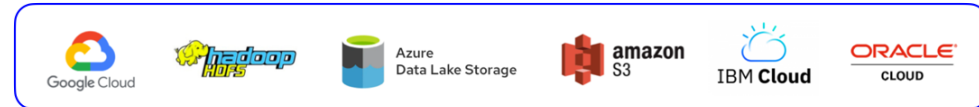
Social Media feed



IOT and Machines Generated Data



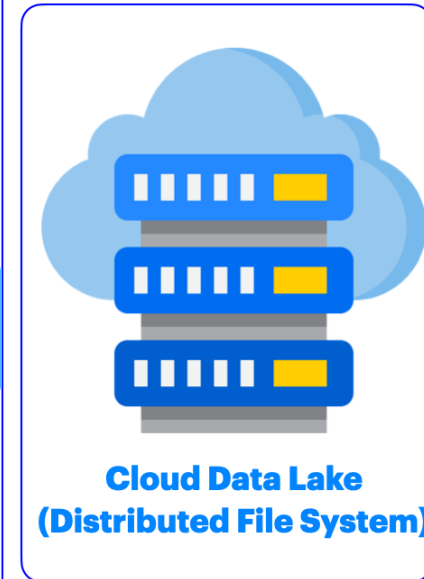
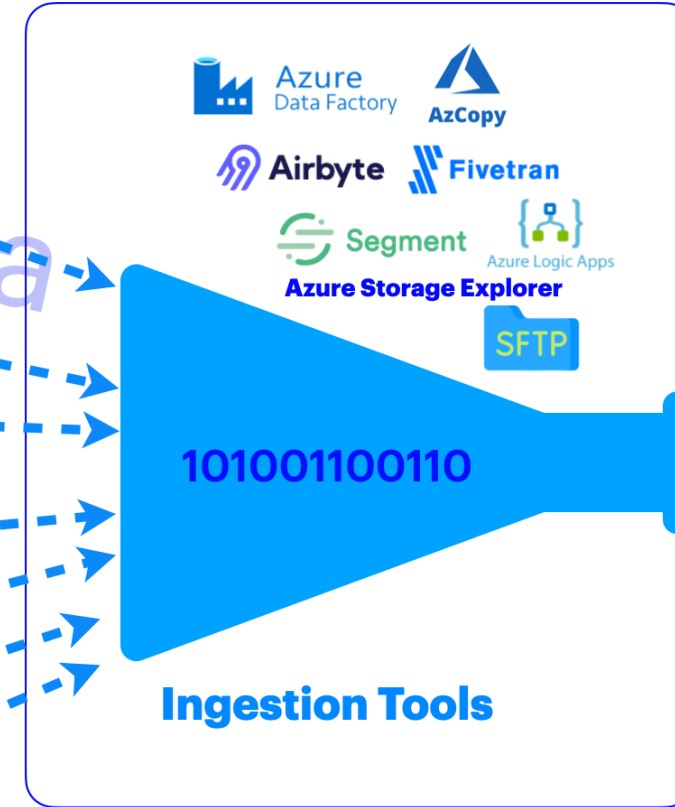
Cloud Data Lakes



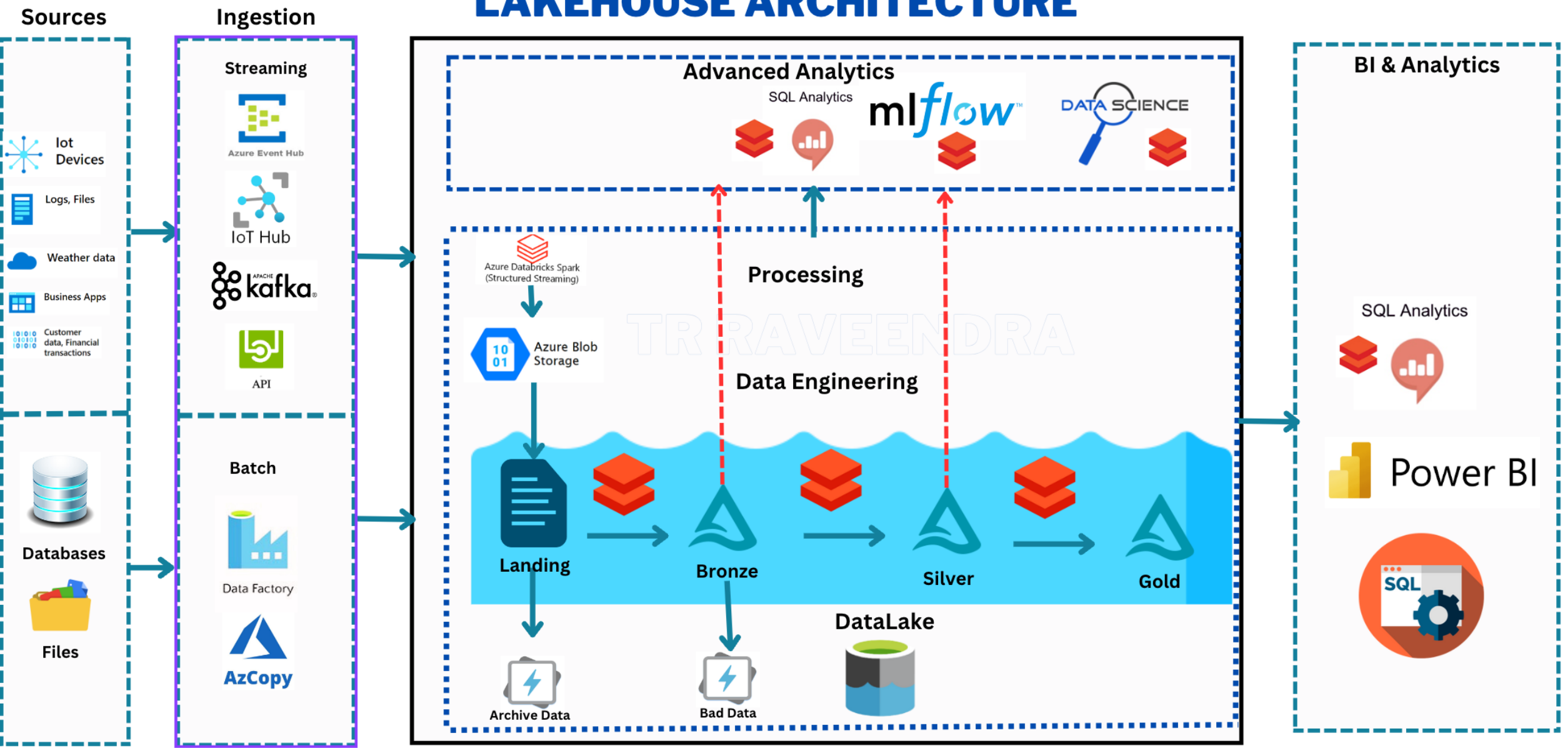
Un-Structured Data



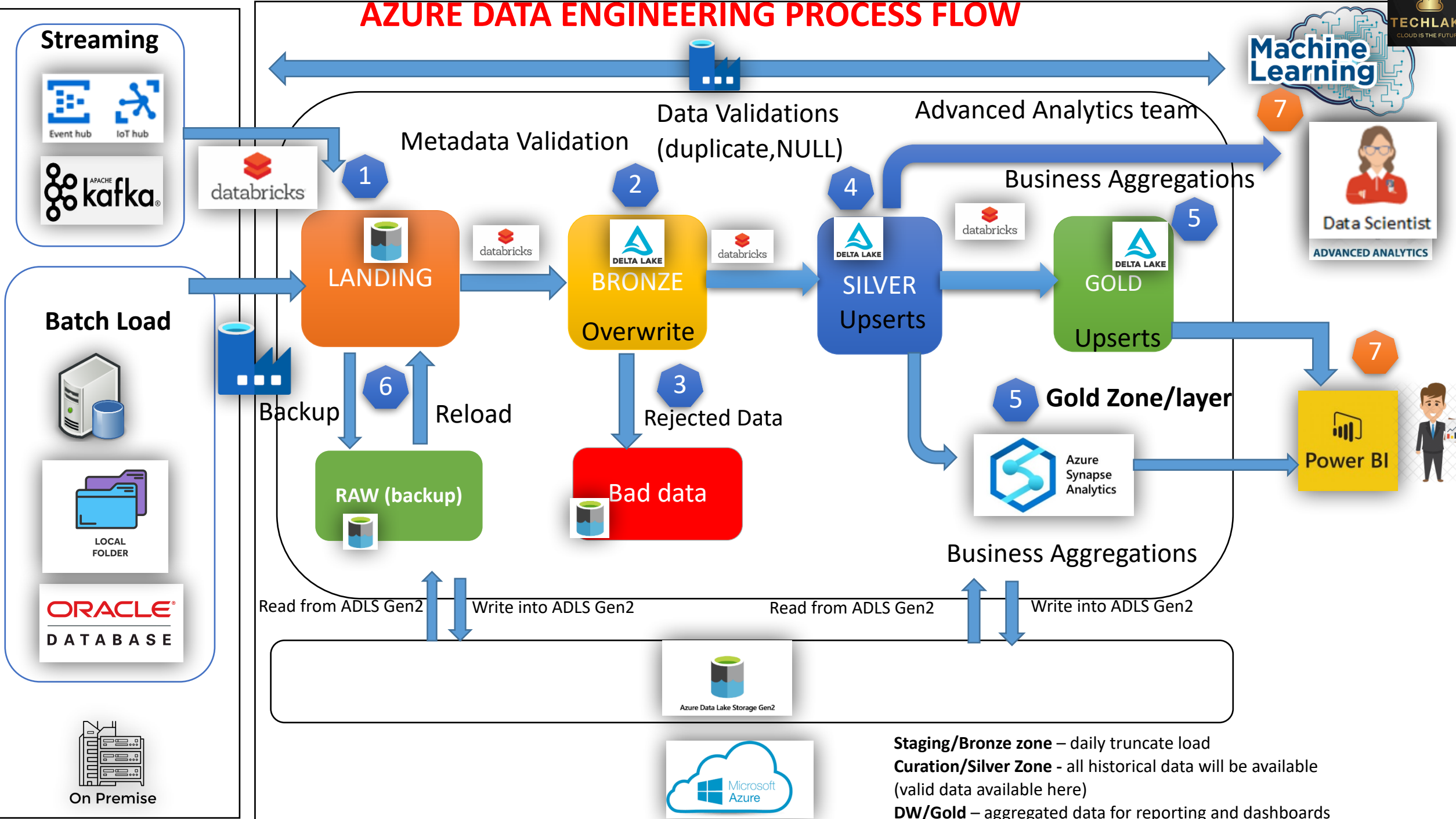
On-Premises - DataBases



LAKEHOUSE ARCHITECTURE



AZURE DATA ENGINEERING PROCESS FLOW



COUNTRY	
Properties	
🔑 COUNTRY_ID	
🔑 COUNTRY_ISO_CODE	
🔑 COUNTRY_NAME	
🔑 COUNTRY_SUBREGION	
🔑 COUNTRY_SUBREGION_ID	
🔑 COUNTRY_REGION	
🔑 COUNTRY_REGION_ID	
🔑 COUNTRY_TOTAL	
🔑 COUNTRY_TOTAL_ID	
🔑 COUNTRY_NAME_HIST	
Navigation Properties	
🔑 CUSTOMERS	

CUSTOMER	
Properties	
🔑 CUST_ID	
🔑 CUST_FIRST_NAME	
🔑 CUST_LAST_NAME	
🔑 CUST_GENDER	
🔑 CUST_YEAR_OF_BIRTH	
🔑 CUST_MARITAL_STATUS	
🔑 CUST_STREET_ADDRESS	
🔑 CUST_POSTAL_CODE	
🔑 CUST_CITY	
🔑 CUST_CITY_ID	
🔑 CUST_STATE_PROVINCE	
🔑 CUST_STATE_PROVINCE_ID	
🔑 COUNTRY_ID	
🔑 CUST_MAIN_PHONE_NUM...	
🔑 CUST_INCOME_LEVEL	
🔑 CUST_CREDIT_LIMIT	
🔑 CUST_EMAIL	
🔑 CUST_TOTAL	
🔑 CUST_TOTAL_ID	
🔑 CUST_SRC_ID	
🔑 CUST_EFF_FROM	
🔑 CUST_EFF_TO	
🔑 CUST_VALID	
Navigation Properties	
🔑 COUNTRY	
🔑 SALES	

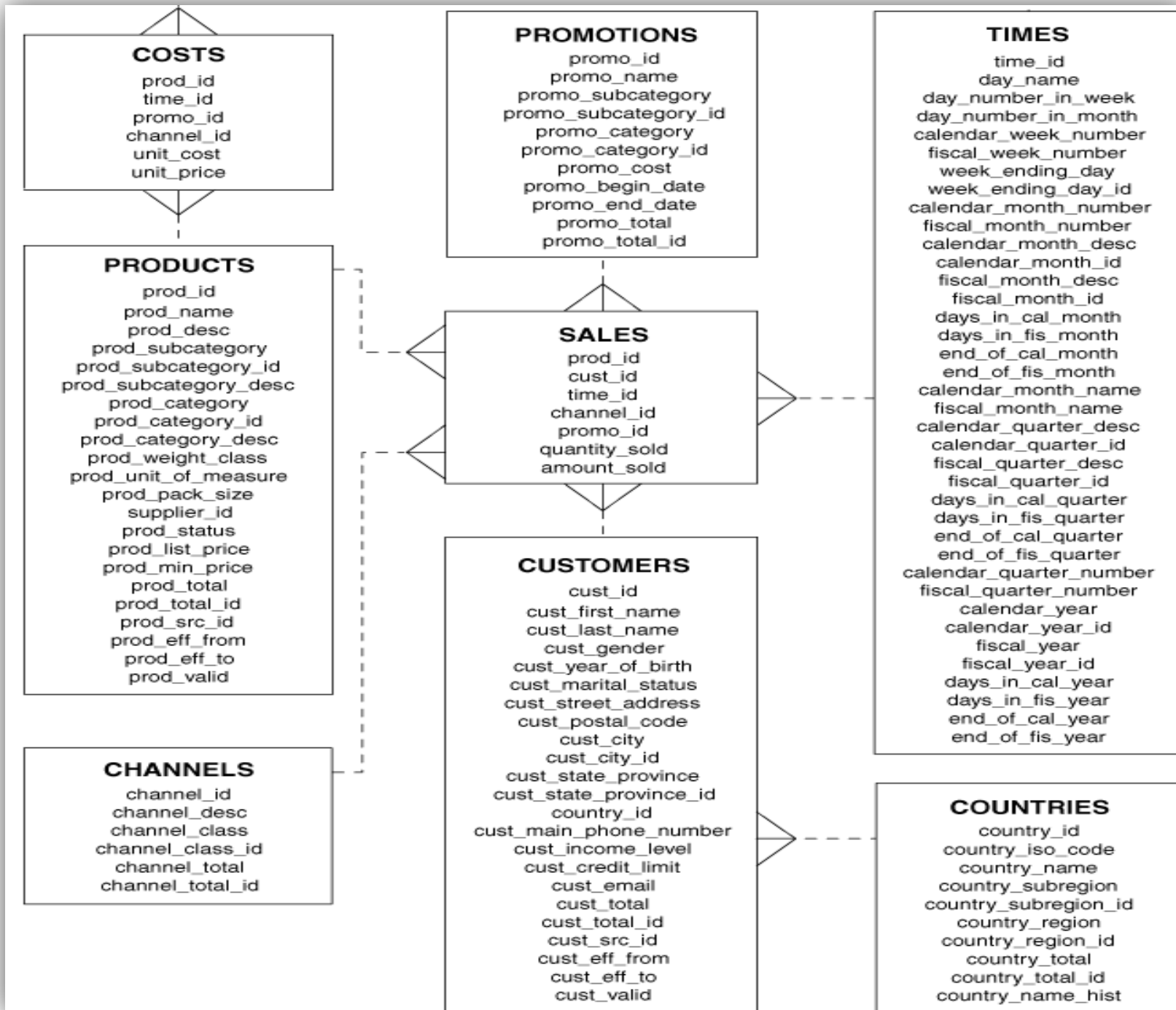
SALE	
Properties	
🔑 PROD_ID	
🔑 CUST_ID	
🔑 TIME_ID	
🔑 CHANNEL_ID	
🔑 PROMO_ID	
🔑 QUANTITY_SOLD	
🔑 AMOUNT_SOLD	
Navigation Properties	
🔑 CHANNEL	
🔑 CUSTOMER	
🔑 PRODUCT	
🔑 PROMOTION	
🔑 TIME	

PROMOTION	
Properties	
🔑 PROMO_ID	
🔑 PROMO_NAME	
🔑 PROMO_SUBCATEGORY	
🔑 PROMO_SUBCATEGORY_ID	
🔑 PROMO_CATEGORY	
🔑 PROMO_CATEGORY_ID	
🔑 PROMO_COST	
🔑 PROMO_BEGIN_DATE	
🔑 PROMO_END_DATE	
🔑 PROMO_TOTAL	
🔑 PROMO_TOTAL_ID	
Navigation Properties	
🔑 SALES	

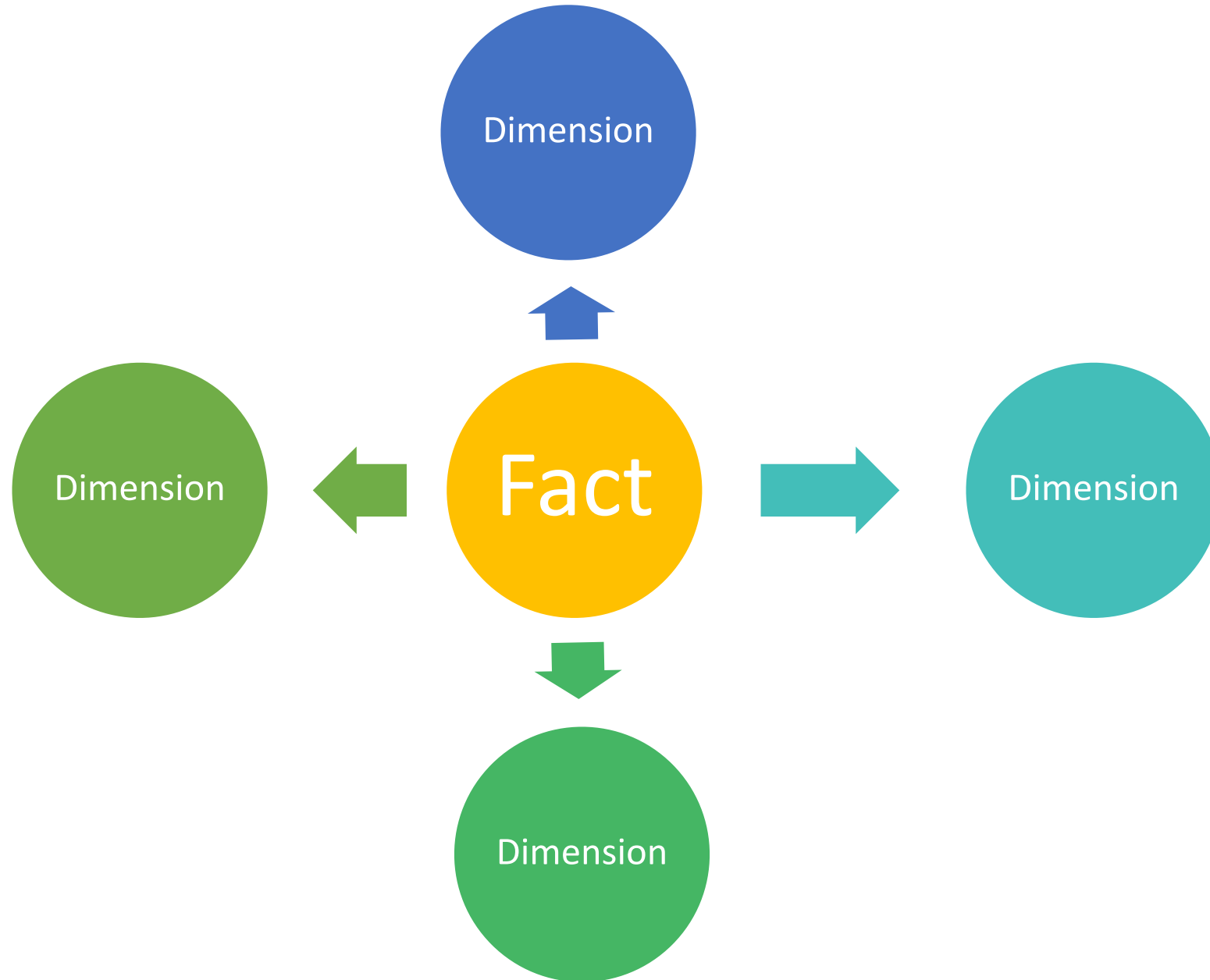
TIME	
Properties	
🔑 TIME_ID	
🔑 DAY_NAME	
🔑 DAY_NUMBER_IN_WEEK	
🔑 DAY_NUMBER_IN_MONTH	
🔑 CALENDAR_WEEK_NUMBER	
🔑 FISCAL_WEEK_NUMBER	
🔑 WEEK_ENDING_DAY	
🔑 WEEK_ENDING_DAY_ID	
🔑 CALENDAR_MONTH_NUMBER	
🔑 FISCAL_MONTH_NUMBER	
🔑 CALENDAR_MONTH_DESC	
🔑 CALENDAR_MONTH_ID	
🔑 FISCAL_MONTH_DESC	
🔑 FISCAL_MONTH_ID	
🔑 DAYS_IN_CAL_MONTH	
🔑 DAYS_IN_FIS_MONTH	
🔑 END_OF_CAL_MONTH	
🔑 END_OF_FIS_MONTH	
🔑 CALENDAR_MONTH_NAME	
🔑 FISCAL_MONTH_NAME	
🔑 CALENDAR_QUARTER_DESC	
🔑 CALENDAR_QUARTER_ID	
🔑 FISCAL_QUARTER_DESC	
🔑 FISCAL_QUARTER_ID	
🔑 DAYS_IN_CAL_QUARTER	
🔑 DAYS_IN_FIS_QUARTER	
🔑 END_OF_CAL_QUARTER	
🔑 END_OF_FIS_QUARTER	
🔑 CALENDAR_QUARTER_NUMBER	
🔑 FISCAL_QUARTER_NUMBER	
🔑 CALENDAR_YEAR	
🔑 CALENDAR_YEAR_ID	
🔑 FISCAL_YEAR	
🔑 FISCAL_YEAR_ID	
🔑 DAYS_IN_CAL_YEAR	
🔑 DAYS_IN_FIS_YEAR	
🔑 END_OF_CAL_YEAR	
🔑 END_OF_FIS_YEAR	
Navigation Properties	
🔑 SALES	

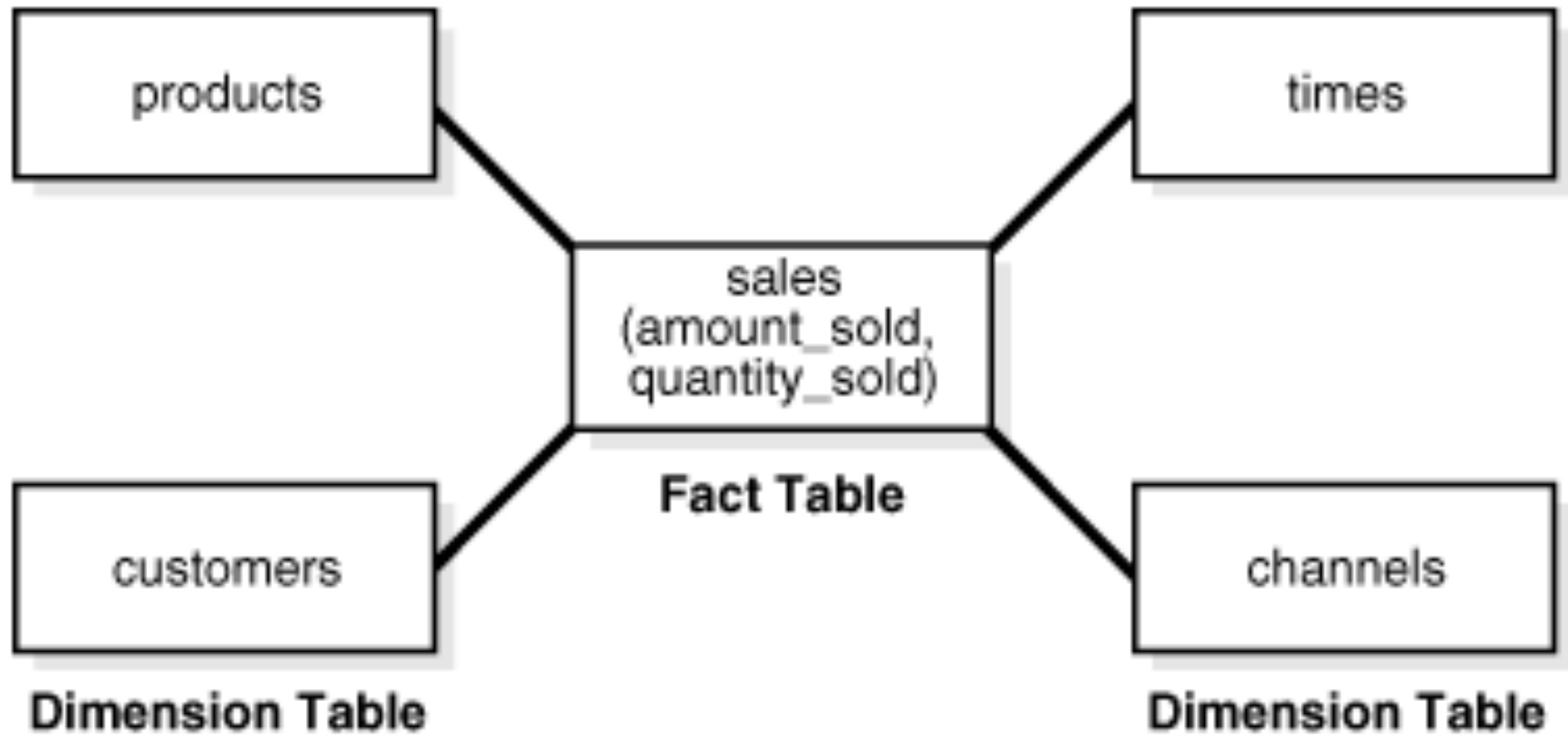
CHANNEL	
Properties	
🔑 CHANNEL_ID	
🔑 CHANNEL_DESC	
🔑 CHANNEL_CLASS	
🔑 CHANNEL_CLASS_ID	
🔑 CHANNEL_TOTAL	
🔑 CHANNEL_TOTAL_ID	
Navigation Properties	
🔑 SALES	

PRODUCT	
Properties	
🔑 PROD_ID	
🔑 PROD_NAME	
🔑 PROD_DESC	
🔑 PROD_SUBCATEGORY	
🔑 PROD_SUBCATEGORY_ID	
🔑 PROD_SUBCATEGORY_DESC	
🔑 PROD_CATEGORY	
🔑 PROD_CATEGORY_ID	
🔑 PROD_CATEGORY_DESC	
🔑 PROD_WEIGHT_CLASS	
🔑 PROD_UNIT_OF_MEASURE	
🔑 PROD_PACK_SIZE	
🔑 SUPPLIER_ID	
🔑 PROD_STATUS	
🔑 PROD_LIST_PRICE	
🔑 PROD_MIN_PRICE	
🔑 PROD_TOTAL	
🔑 PROD_TOTAL_ID	
🔑 PROD_SRC_ID	
🔑 PROD_EFF_FROM	
🔑 PROD_EFF_TO	
🔑 PROD_VALID	
Navigation Properties	
🔑 SALES	

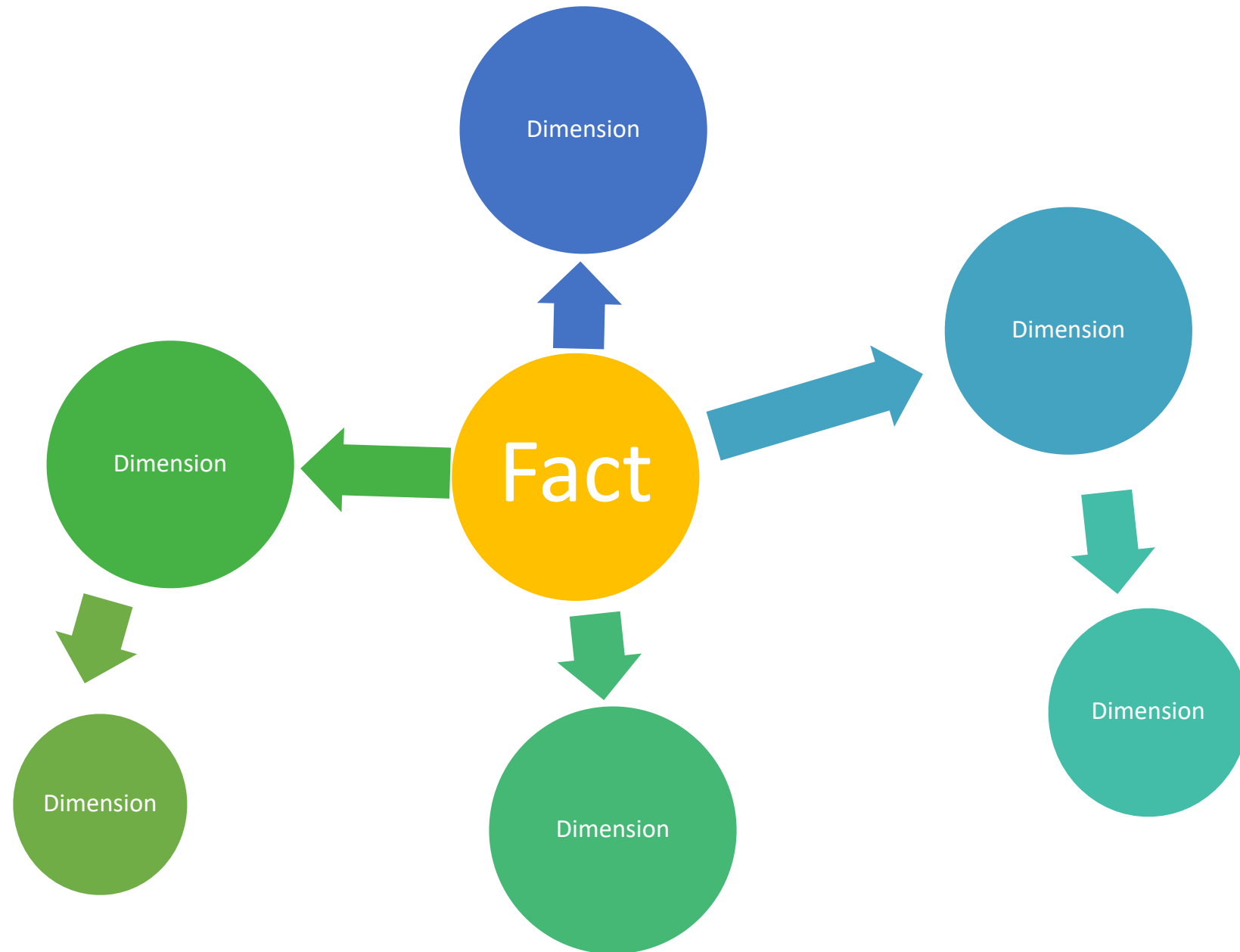


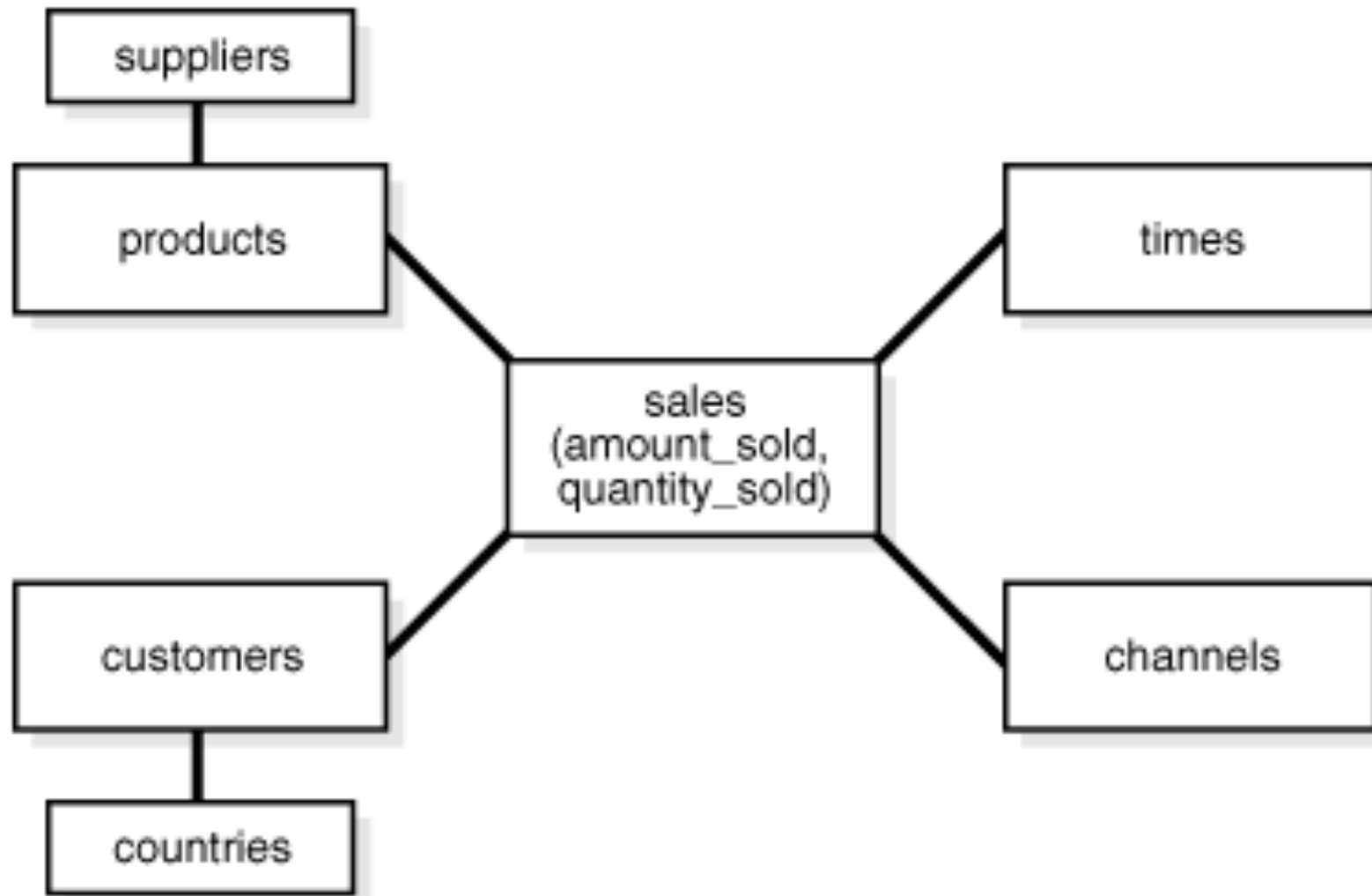
Star Scheme Design



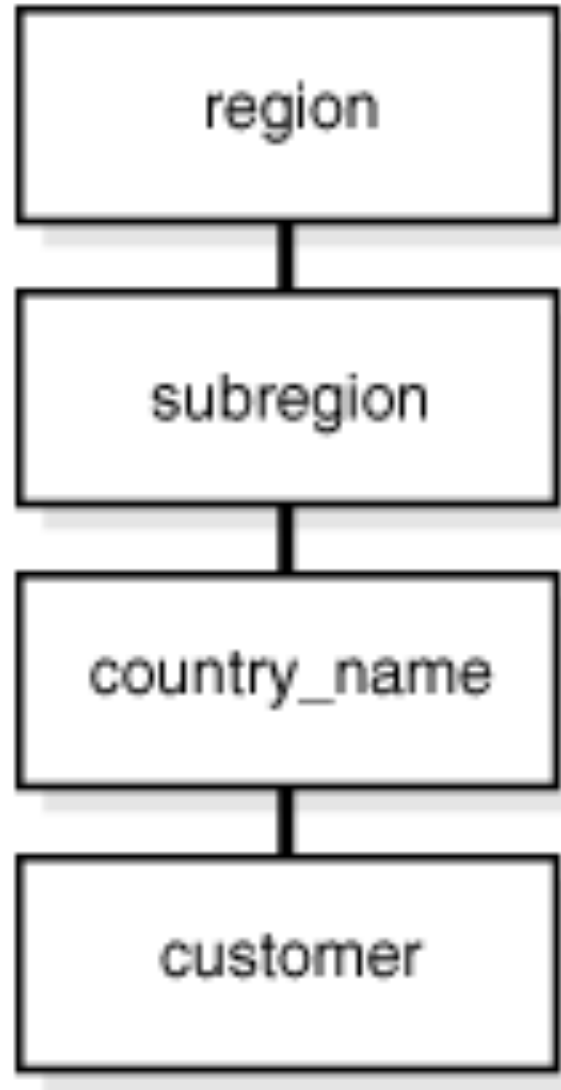


Snowflake Scheme Design





Typical Levels in a Dimension Hierarchy



How to Create Scope in Databricks to access Secrets from Key Vault

← → ↺ 🔒 adb-3784951373135514.14.azure.databricks.net/?o=3784951373135514#secrets/createScope

Apps What's New in Azur... Script Download Li... Azure Data Factory... Beautiful, Free peo... Today's Famous Birt... Today in I

Microsoft Azure | Databricks

HomePage / Create Secret Scope

Create Secret Scope |

A store for secrets that is identified by a name and backed by a specific store type. [Learn more](#)

Scope Name [?]

batch19-adlsgen2-scope

Manage Principal [?]

Creator

Azure Key Vault [?]

DNS Name

https://azurekv-ravi.vault.azure.net/

Resource ID

/subscriptions/e0bf1971-b822-46ca-8e4a-72555f0a959e/resourceGroups/rg-de-dev

1/3

How to Create Scope in Databricks to access Secrets from Key Vault

azurekv-ravi | Properties

Key vault

Save Discard changes Refresh

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Events
- Settings
 - Keys
 - Secrets
 - Certificates
 - Access policies
 - Networking
 - Security
 - Properties**

Name	azurekv-ravi
Sku (Pricing tier)	Standard
Location	eastus
Vault URI	https://azurekv-ravi.vault.azure.net/
Resource ID	/subscriptions/e0bf1971-b822-46ca-8e4a-72555f0a959e/resourceGroups/rg-de-dev/providers/
Subscription ID	e0bf1971-b822-46ca-8e4a-72555f0a959e
Subscription Name	Pay-As-You-Go
Directory ID	89556edd-0e4a-43c1-bbcb-9139ff026b1f
Directory Name	Default Directory
Soft-delete	Soft delete has been enabled on this key vault
Days to retain deleted vaults	90
Purge protection	☒ Disable purge protection (allow key vault and objects to be purged during retention period)

Cmd 9

```

1 # HDFS / DBFS / GFS / DFS / WASBS - windows azure storage blob service / AWS S3 - simple storage Service / DBFS (Databricks file system)
2 dbutils.fs.mount(
3     source = "wasbs://batch19@adlsgen2batch19.blob.core.windows.net/source/landing/sales",
4     mount_point = "/mnt/landing/sales",
5     extra_configs = {"fs.azure.account.key.adlsgen2batch19.blob.core.windows.net":dbutils.secrets.get(scope="batch19-adlsgen2-scope",key="batch19-adlsgen2-account-key")})

```

► (1) Spark Jobs

Out[2]: True

Master Pipeline

pl_master_sales_load

Activities

Search activities

Move & transform

Azure Data Explorer

Azure Function

Batch Service

Databricks

Data Lake Analytics

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Save as template

Validate

Debug

Add trigger

foreach for processing one by one job from control table and calling child pipeline

Lookup

Lookup_Getjobdetails

ForEach

ForEach_Eachjob

Activities

1 activities

Execute Pipeline

pl_alerts

Sending Email Alerts after processing

Parameters

Variables

Settings

Output

Pipeline run ID:

Name	Type	Run start	Duration	Status	Integr
------	------	-----------	----------	--------	--------

Lookup Activity

pl_master_sales_load

Activities

Search activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Lookup

Lookup_Getjobdetails

ForEach

ForEach_Eachjob

Activities
1 activities

Execute Pipeline

pl_alerts

General **Settings** User properties

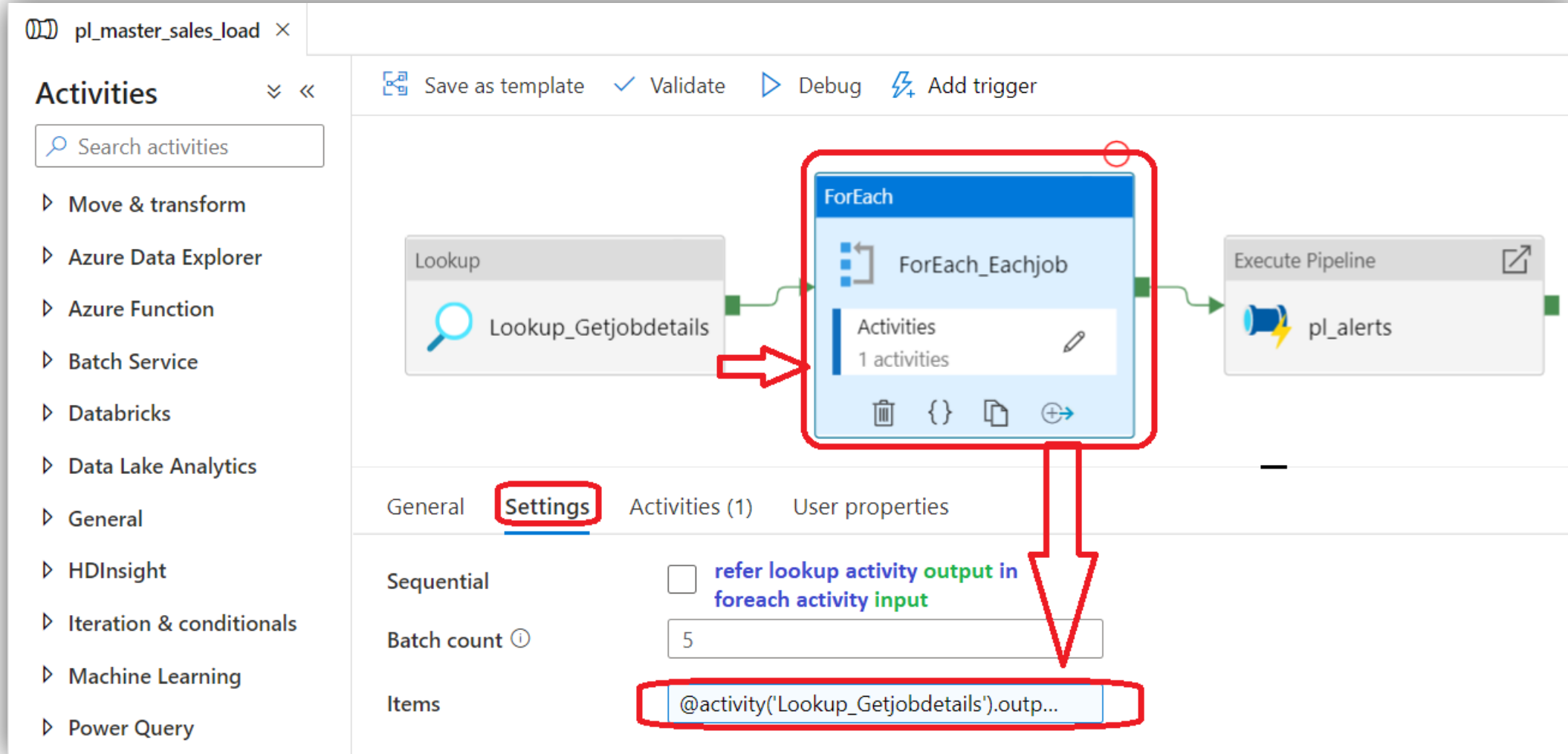
Source dataset * ds_deltalake_lookup

Use query ☐ Table ☒ Query

Query *
select * from jobs.JOB_LIST_DYNAMIC
order by job_id

First row only ☐

ForEach Activity



The screenshot shows the Azure Data Factory interface for a pipeline named 'pl_master_sales_load'. The pipeline is composed of three activities: 'Lookup_Getjobdetails', 'ForEach_Eachjob', and 'Execute Pipeline pl_alerts'. The 'ForEach_Eachjob' activity is highlighted with a red box. A red arrow points from the 'Lookup_Getjobdetails' activity to the 'ForEach_Eachjob' activity. Another red arrow points from the 'Items' field in the 'Settings' tab of the 'ForEach_Eachjob' activity to the text '@activity('Lookup_Getjobdetails').output.value'.

Activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

ForEach Activity Settings

Property	Value
Sequential	☐ refer lookup activity output in foreach activity input
Batch count	5
Items	@activity('Lookup_Getjobdetails').output.value

@activity('Lookup_Getjobdetails').output.value

Calling Child pipeline in Master Pipeline ForEach Activity to process one by one

Activities

- Move & transform
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- Data Lake Analytics
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

pl_master_sales_load > ForEach_Eachjob

Execute Pipeline

pl_child_sales_load

calling child pipeline in master pipeline inside foreach loop

Settings

Invoked pipeline * **pl_child_sales_load** Open + New

Wait on completion ☒

Parameters

specify all parameter values in pipeline and get values from foreach **@item().parameter**

Name	Type	Value
job_id	string	@item().job_id
job_name	string	@item().job_name
source_folder_name	string	@item().source_folder_name
landing_zone_file_path	string	@item().landing_zone_file_path

Chile Pipeline with Parameters

pl_master_sales_load

pl_child_sales_load

×

Save as template Validate Debug Add trigger

LookupStartLogging

Get MetadataGetMetadata_Onprem

FilterFilter_Files

ForEachForEach_CopyFile

1 activities

NotebookLandingToStaging

WaitWait1

NotebookStagingToCuration

LookupLandingToStaging_Log

WaitWait

LookupStagingToCuration_Log

If ConditionIfDim

True2 activities

FalseNo activities

SwitchSwitchFacts

DefaultNo activities

sales_transaction1 activities

costs_transaction1 activities

ParametersVariablesSettingsOutput

+ New

Delete

☐ Name	Type	Default value
☐ job_id	String	Value
☐ job_name	String	Value
☐ source folder name	String	Value

pl_child_sales_load

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Parameters

Variables

Settings

Output

☐	Name	Type	Default value
☐	job_id	String	Value
☐	job_name	String	Value
☐	source_folder_name	String	Value
☐	landing_zone_file_path	String	Value
☐	landing_zone_folder_name	String	Value
☐	landing_zone_file_name	String	Value
☐	staging_zone_database_name	String	Value
☐	staging_zone_table_name	String	Value
☐	staging_zone_table_pk_column	String	Value
☐	curation_zone_database_name	String	Value
☐	curation_zone_table_name	String	Value
☐	curation_zone_table_pk_column	String	Value
☐	dw_zone_database_name	String	Value
☐	dw_zone_table_name	String	Value
☐	dw_zone_table_pk_column	String	Value
☐	pyspark_schema	String	Value
☐	table_type	String	Value
☐	raw_zone_file_path	String	Value
☐	raw_zone_folder_name	String	Value
☐	raw_zone_file_name	String	Value

TECHLAKE
CLOUD IS THE FUTURE.

pl_child_sales_load ×

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Lookup

StartLogging

Get Metadata

GetMetadata_Onprem

Filter

Filter_Files

General **Settings** User properties

Source dataset * ds_deltalake_lookup Open + New Preview data

Use query ☐ Table ☒ Query

Query insert into JOBS.pipeline_log select sh...

First row only ☒

```
insert into JOBS.pipeline_log select  
sha2('@{pipeline().RunId}',256),@{pipeline().parameters.job_id},@{pipeline().DataFactory},'@{pipeline().Pipeline}','@{pipeline().RunId}','@{pipeline().TriggerType}','@{pipeline().TriggerId}','  
@{pipeline().TriggerName}','@{pipeline().TriggerTime}',"","","","","","","","","","","","","","","","",""
```

Child pipeline – GetMetadata Activity to get files from Onprem

pl_child_sales_load

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Lookup StartLogging → **Get Metadata GetMetadata_Onprem** → Filter Filter_Files

General **Dataset** User properties

Dataset * ds_onprem_file_getmetadata  Open  New [Learn more](#) 

Dataset properties ⓘ

Name	Value
ds_folder_name	@pipeline().parameters.source_folder_...

Filter by last modified ⓘ

Start time (UTC) End time (UTC)

Field list  New |  Delete

- ☐ Argument
- ☐ Child items 
- ☐ Item name 

Dataset for GetMetadata from onprem

pl_master_sales_load pl_child_sales_load **ds_onprem_file_get...**

01 Binary
ds_onprem_file_getmetadata

Connection **Parameters**

+ New | Delete

Name	Type	Default value
☐ ds_folder_name	String	Value

pl_master_sales_load pl_child_sales_load **ds_onprem_file_get...**

01 Binary
ds_onprem_file_getmetadata

Connection Parameters

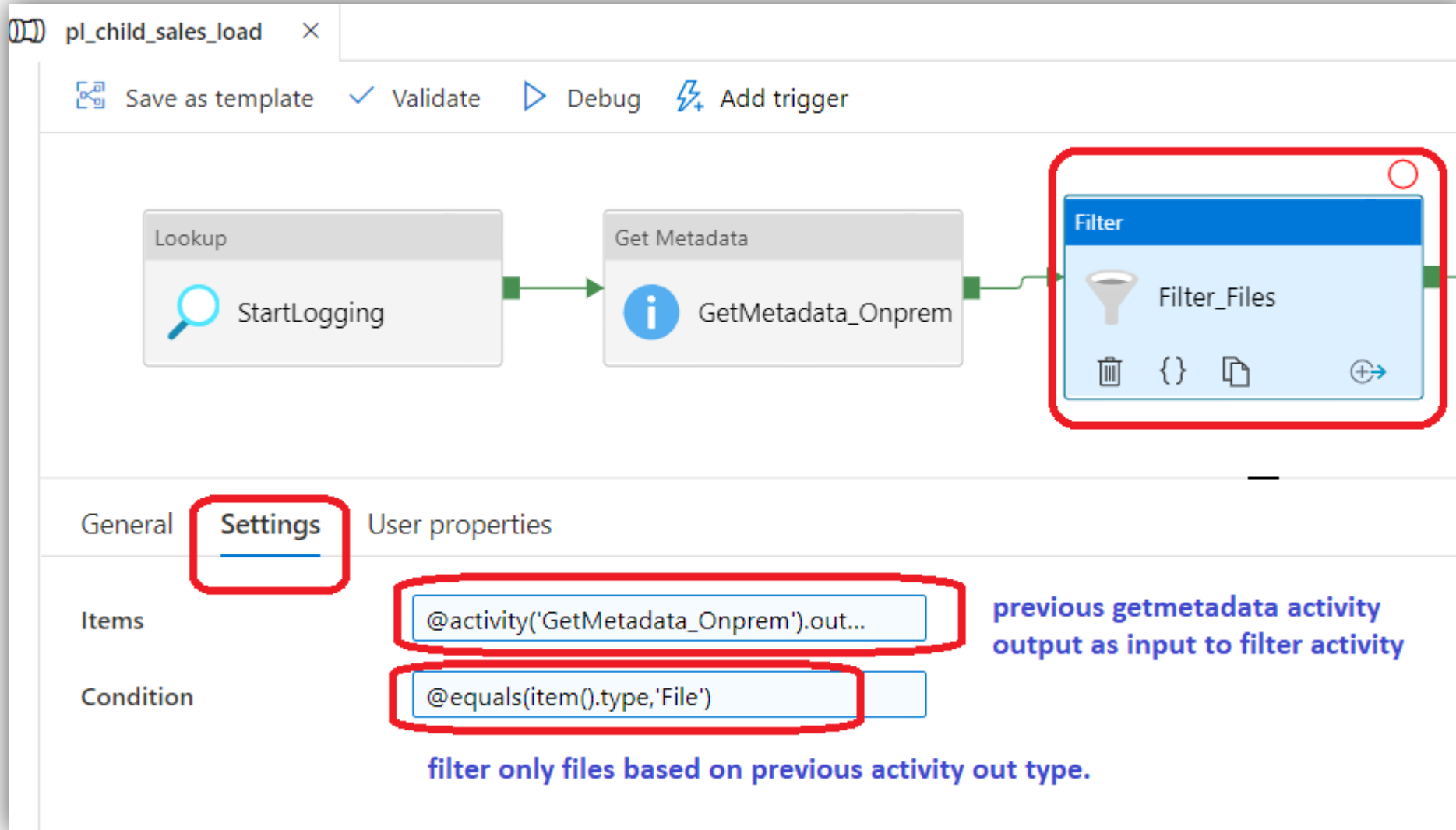
Linked service * ls_file_onprem_source Test connection Edit + New

Integration runtime * OIRSelfhosted (Unavailable) Edit

File path * E:\files\SH / **@dataset().ds_folder_name** / File

Compression type None

Filter-Activity For filter only files.



The screenshot displays the Azure Data Factory (ADF) interface for a pipeline named `pl_child_sales_load`. The workflow consists of three activities: `Lookup` (containing `StartLogging`), `Get Metadata` (containing `GetMetadata_Onprem`), and `Filter` (containing `Filter_Files`). The `Filter` activity is highlighted with a red box. Below the workflow, the `Settings` tab is selected and also highlighted with a red box. The `Items` property is set to `@activity('GetMetadata_Onprem').out...`, and the `Condition` is set to `@equals(item().type,'File')`. Both the `Items` and `Condition` expressions are highlighted with red boxes. Explanatory text in blue states: "previous getmetadata activity output as input to filter activity" and "filter only files based on previous activity out type."

pl_child_sales_load

Save as template Validate Debug Add trigger

Lookup
StartLogging

Get Metadata
GetMetadata_Onprem

Filter
Filter_Files

General **Settings** User properties

Items
Condition

`@activity('GetMetadata_Onprem').out...`

`@equals(item().type,'File')`

previous getmetadata activity output as input to filter activity

filter only files based on previous activity out type.

Foreach Activity

Calling COPY Activity to copy files from on-prem to azure

pl_child_sales_load

Save as template Validate Debug Add trigger

foreach activity to copy files from onprem to azure

Filter

Filter_Files

ForEach

ForEach_CopyFile

Activities

1 activities

Notebook

LandingToStaging

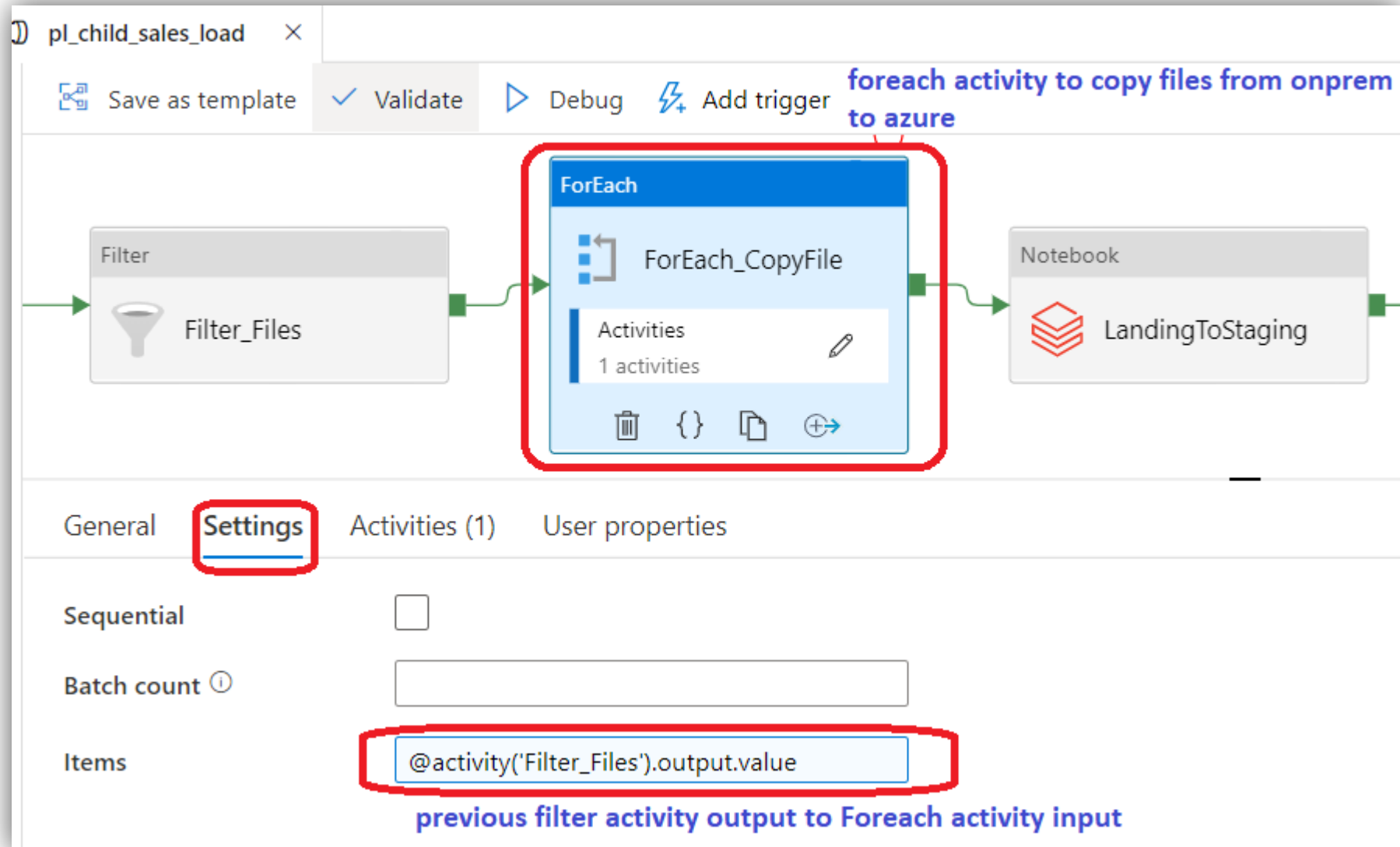
General Settings Activities (1) User properties

Sequential ☐

Batch count

Items

previous filter activity output to Foreach activity input



Copy- Activity

pl_child_sales_load ×

Save as template ✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

pl_child_sales_load > ForEach_CopyFile

Copy activity for processing files from onprem to azure.
Source: On-prem file
Target: azure Data lake adls gen2

Copy data

OnPremToLanding

General **Source** Sink Mapping Settings User properties

Source dataset * ds_onprem_file_binary Open + New Learn more

Dataset properties

Name	Value	Type
ds_folder_name	@pipeline().parameters.source_folder_...	string
ds_file_name	@item().name	string

File path type ☒ File path in dataset ☐ File filter ☐ Wildcard file path ☐ List of files

Start time (UTC) End time (UTC)

Filter by last modified

Recursively ☒

Copy-Activity

Source Dataset - Properties

pl_master_sales_load pl_child_sales_load **ds_onprem_file_binary**

01 Binary
ds_onprem_file_binary

Connection **Parameters**

+ New | Delete

☐ Name	Type	Default value
☐ ds_folder_name	String	Value
☐ ds_file_name	String	Value

pl_master_sales_load pl_child_sales_load **ds_onprem_file_binary**

01 Binary
ds_onprem_file_binary

Connection Parameters

Linked service * ls_file_onprem_source Test connection Edit + New Learn more

Integration runtime * OIRSelfhosted (Unavailable) Edit

File path * E:\files\SH / @dataset().ds_folder_name / @dataset().ds_file_name

Compression type None

Copy-Activity

Sink- Dataset properties

pl child sales load

Save as template ✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

pl child sales load > ForEach CopyFile

Copy data

OnPremToLanding

General Source **Sink** Mapping Settings User properties

Sink dataset * ds_adlsgen2_landing_binary

Dataset properties ⓘ

Name	Value
ds_folder_name	@pipeline().parameters.landing_zone_...
ds_file_name	@item().name

Copy behavior ⓘ None

Sink DataSet Parameters and dynamic filepath.

pl_master_sales_load pl_child_sales_load ds_adlsgen2_landing... X

01 Binary
ds_adlsgen2_landing_binary

Connection **Parameters**

+ New | Delete

☐ Name	Type	Default value
☐ ds_folder_name	String	Value
☐ ds_file_name	String	Value

pl_master_sales_load pl_child_sales_load ds_adlsgen2_landing... X

01 Binary
ds_adlsgen2_landing_binary

Connection Parameters

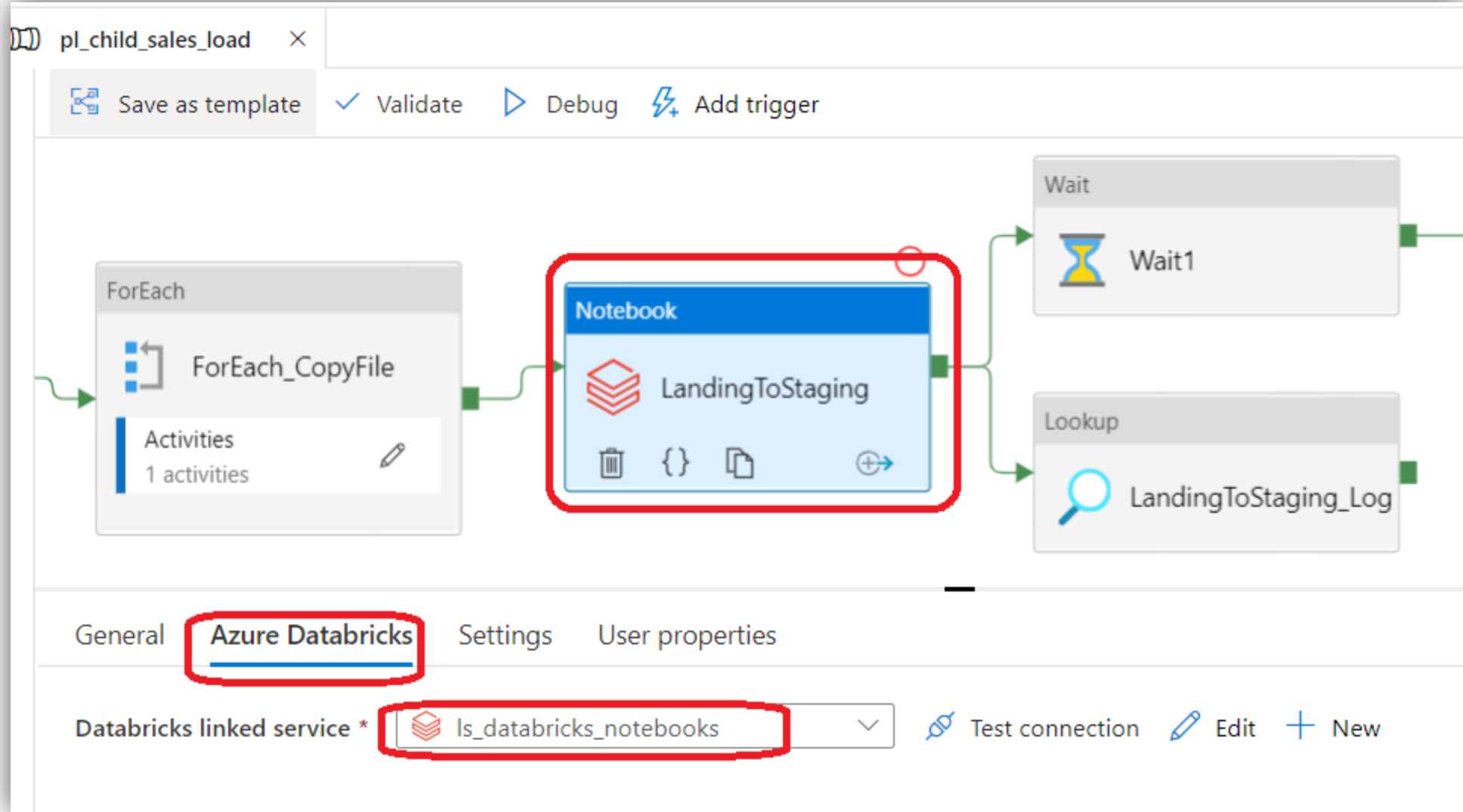
Linked service * Is_adlsgen2_sales Test connection Edit + New Learn more

File path * sales / @concat('landing/sales/',dataset().ds_f... / @dataset().ds_file_name

Compression type None

@concat('landing/sales/',dataset().ds_folder_name)

Calling Databricks Notebook activity for Landing to Staging.



Notebook : Landing to staging widgets parameters

pl_child_sales_load

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

ForEach

ForEach_CopyFile

Activities
1 activities

Notebook

LandingToStaging

Wait

Wait1

Lookup

LandingToStaging_Log

General Azure Databricks **Settings** User properties

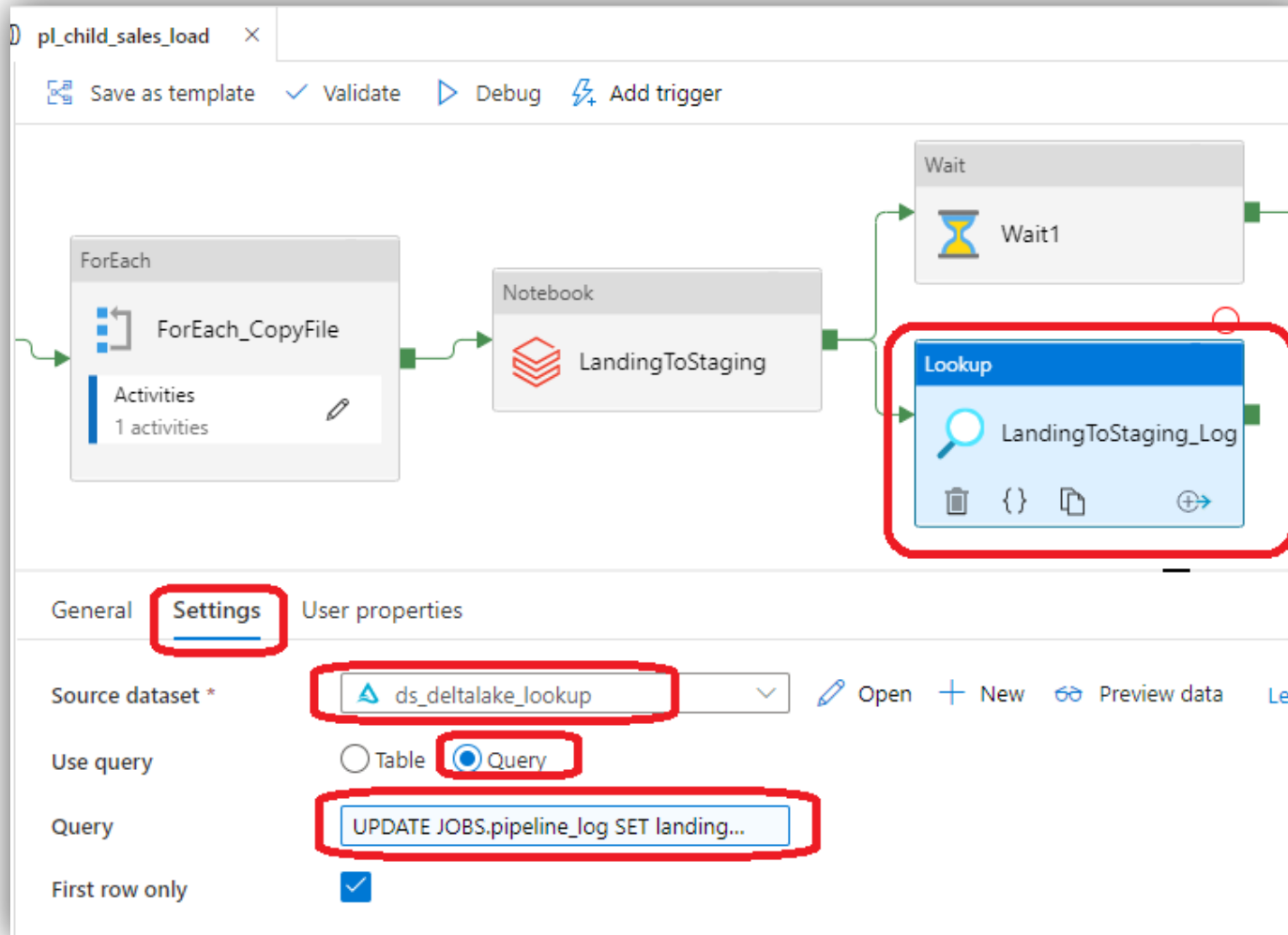
Notebook path * Browse Open

Base parameters

+ New | Delete

☐ Name	Value
☐ landing_zone_file_pat	@pipeline().parameters.landing_zone_...
☐ landing_zone_folder_r	@pipeline().parameters.landing_zone_...
☐ landing_zone_file_nam	@pipeline().parameters.landing_zone_...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...
☐ pyspark_schema	@pipeline().parameters.pyspark_schema
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id

Lookup- Activity for logging into delta lake table



The screenshot shows the configuration of a Lookup activity named 'LandingToStaging_Log' within a pipeline named 'pl_child_sales_load'. The pipeline flow is as follows:

- ForEach** activity (containing 'ForEach_CopyFile') connects to the **Notebook** activity 'LandingToStaging'.
- The **Notebook** activity connects to the **Lookup** activity 'LandingToStaging_Log'.
- The **Lookup** activity connects to a **Wait** activity 'Wait1'.

The **Lookup** activity configuration is shown in the bottom panel:

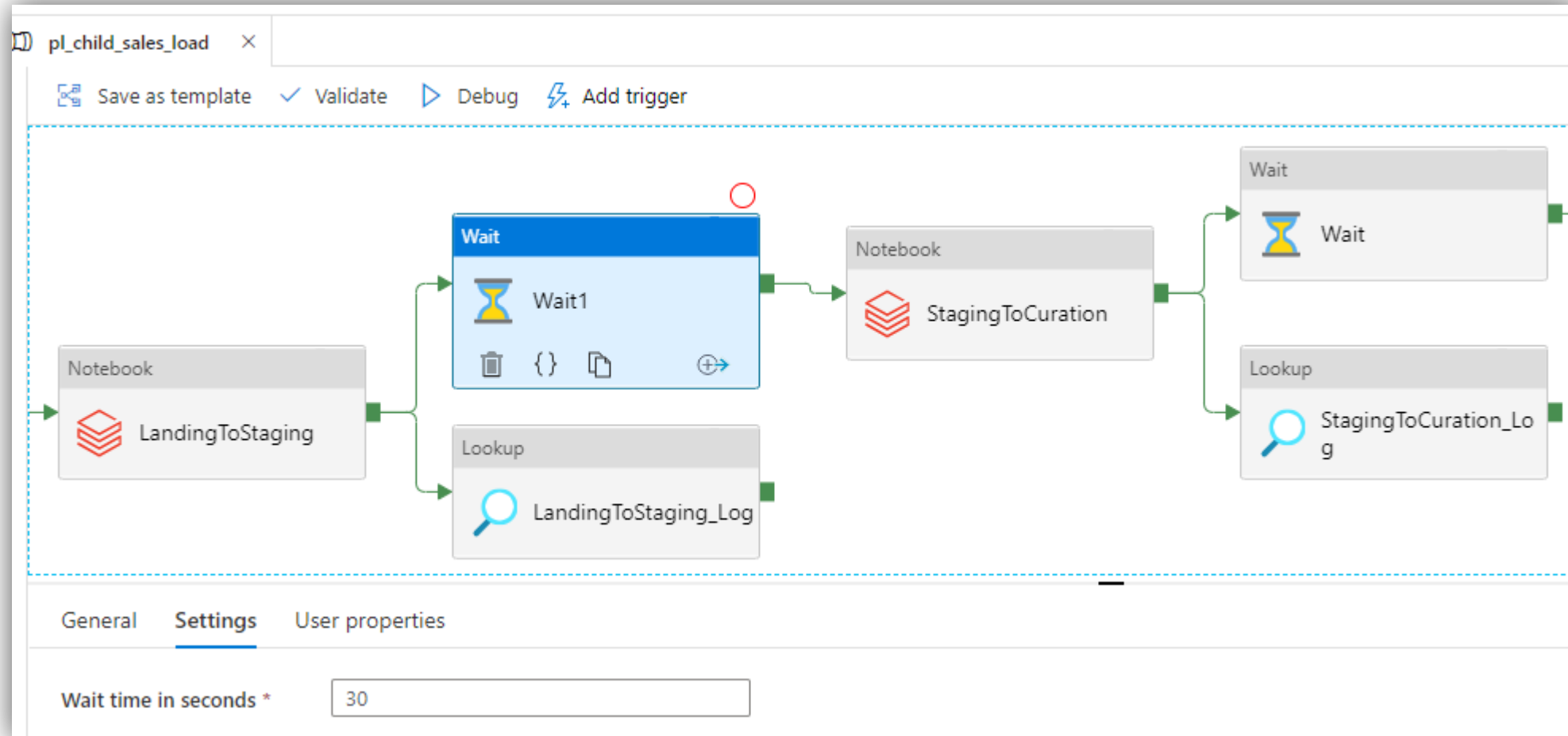
- Source dataset ***: ds_deltalake_lookup
- Use query**: ☒ Query
- Query**: UPDATE JOBS.pipeline_log SET landing...
- First row only**: ☒

UPDATE JOBS.pipeline_log SET

landing_rows=@{activity('LandingToStaging').output.runOutput.landingRows},rejected_rows=@{activity('LandingToStaging').output.runOutput.rejectedRows},
staging_rows=@{activity('LandingToStaging').output.runOutput.stagingRows} WHERE pipeline_runid='@{pipeline().RunId}'

Wait- Activity

Wait activity to wait job execution few seconds (30 sec) before staging next notebook



Notebook : Activity for Staging to Curation

pl_child_sales_load x

Save as template ✓ Validate ▶ Debug ⚡ Add trigger

Wait
Wait1

Lookup
LandingToStaging_Log

Notebook
StagingToCuration

Wait
Wait

Lookup
StagingToCuration_Log

General Azure Databricks **Settings** User properties

Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

▲ Base parameters

+ New | Delete

☐ Name	Value
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id
☐ curation_zone_databa	@pipeline().parameters.curation_zone...
☐ curation_zone_table_n	@pipeline().parameters.curation_zone...
☐ curation_zone_table_p	@pipeline().parameters.curation_zone...

Lookup- Activity for updating log

The screenshot displays the Azure Data Factory pipeline editor for a pipeline named `pl_child_sales_load`. The pipeline flow includes a `Wait` activity, a `Lookup` activity named `LandingToStaging_Log`, a `Notebook` activity named `StagingToCuration`, and another `Wait` activity. A second `Lookup` activity, named `StagingToCuration_Log`, is highlighted with a red rectangle. Below the pipeline canvas, the `Settings` tab is selected, and its configuration is also highlighted with a red rectangle. The settings include:

- Source dataset ***: `ds_deltalake_lookup`
- Use query**: ☒ Query
- Query**: `UPDATE JOBS.pipeline_log SET curatio...`
- First row only**: ☒

UPDATE JOBS.pipeline_log SET
curation_read_rows=@{activity('StagingToCuration').output.runOutput.curationRows['num_affected_rows']},curation_inserted_rows=@{activity('StagingToCuration').output.runOutput.curat
ionRows['num_inserted_rows']},curation_updated_rows=@{activity('StagingToCuration').output.runOutput.curationRows['num_updated_rows']} WHERE
pipeline_runid='@{pipeline().RunId}'

IF-Condition : for processing Dimensions based on condition (table_type='DIM')

pl_child_sales_load

Save as template Validate Debug Add trigger

Wait

Lookup

StagingToCuration_Loading

If Condition

IFDim

Case	Activity
True	2 activities
False	No activities

Switch

SwitchFacts

Case	Activity
Default	No activities
sales_transaction	1 activities
costs_transaction	1 activities

General Activities (2) User properties

Expression

@equals(pipeline().parameters.table_t...

@equals(pipeline().parameters.table_type,'dim')

Case	Activity
True	2 Activities
False	No activities

Notebook: Curation To DWH processing inside if condition

pl_child_sales_load

Save as template Validate Debug Add trigger

pl_child_sales_load > IFDim > True activities

Notebook

CurationToDWHDim

Lookup

CurationToDWHDim_Log

General Azure Databricks **Settings** User properties

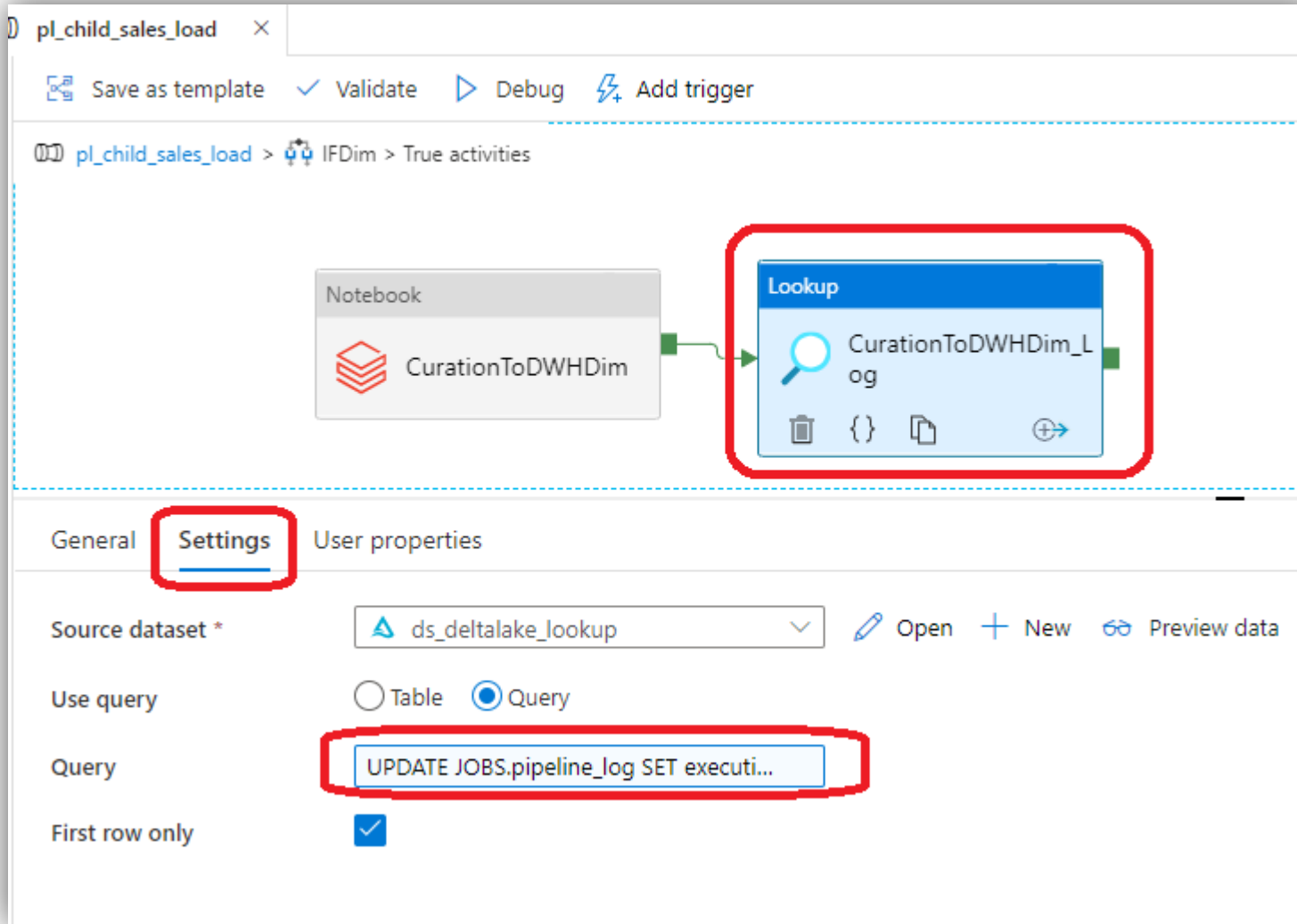
Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

▲ Base parameters

+ New | Delete

☐	Name	Value
☐	job_name	@pipeline().parameters.job_name
☐	job_id	@pipeline().parameters.job_id
☐	curation_zone_databa	@pipeline().parameters.curation_zone...
☐	curation_zone_table_n	@pipeline().parameters.curation_zone...
☐	curation_zone_table_p	@pipeline().parameters.curation_zone...
☐	dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐	dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐	dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...

Updating Log in log table



The screenshot shows the Databricks IDE interface for a pipeline named `pl_child_sales_load`. The pipeline consists of two activities: a `Notebook` activity named `CurationToDWHDim` and a `Lookup` activity. The `Lookup` activity is highlighted with a red box. Below the pipeline, the `Settings` tab is selected, showing the following configuration:

- Source dataset ***: `ds_deltalake_lookup`
- Use query**: ☒ Table ☒ Query
- Query**: `UPDATE JOBS.pipeline_log SET executi...`
- First row only**: ☒

```
UPDATE JOBS.pipeline_log SET
execution_status='SUCCESS',dwh_read_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_affected_rows']],dwh_inserted_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_inserted_rows']],dwh_updated_rows=@{activity('CurationToDWHDim').output.runOutput.dwhRows['num_updated_rows']}} WHERE pipeline_runid=@{pipeline().RunId}'
```

Switch – Case : for processing Fact data loads based on condition.

pl_child_sales_load

Save as template Validate Debug Add trigger

Wait

Lookup

StagingToCuration_Loading

If Condition

IFDim

True

2 activities

False

No activities

Switch

SwitchFacts

Default

No activities

sales_transaction

1 activities

costs_transaction

1 activities

General Activities (2) User properties

Expression

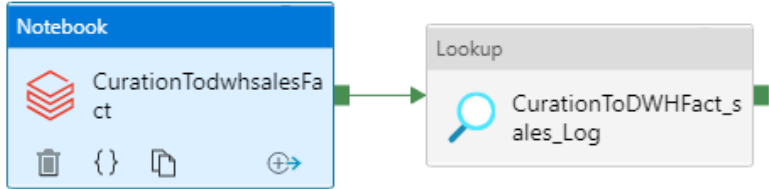
@pipeline().parameters.job_name

+ Add case

Case	Activity
Default	No activities
sales_transaction	1 Activity
costs_transaction	1 Activity

Processing both Facts inside Switch case

pl_child_sales_load > SwitchFacts - salestransaction



General Azure Databricks **Settings** User properties

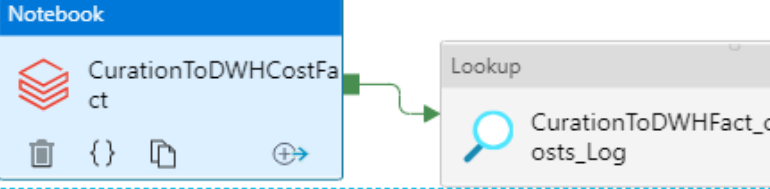
Notebook path * /Shared/Pyspark_Project_Dynamic/NB_sale: Browse Open

▲ Base parameters

+ New | Delete

☐ Name	Value
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id
☐ curation_zone_databa	@pipeline().parameters.curation_zone...
☐ curation_zone_table_n	@pipeline().parameters.curation_zone...
☐ curation_zone_table_p	@pipeline().parameters.curation_zone...
☐ dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐ dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐ dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...

pl_child_sales_load > SwitchFacts - coststransaction



General Azure Databricks **Settings** User properties

Notebook path * /Shared/Pyspark_Project_Dynamic/NB_cost: Browse Open

▲ Base parameters

+ New | Delete

☐ Name	Value
☐ job_name	@pipeline().parameters.job_name
☐ job_id	@pipeline().parameters.job_id
☐ curation_zone_databa	@pipeline().parameters.curation_zone...
☐ curation_zone_table_n	@pipeline().parameters.curation_zone...
☐ curation_zone_table_p	@pipeline().parameters.curation_zone...
☐ dw_zone_database_na	@pipeline().parameters.dw_zone_data...
☐ dw_zone_table_name	@pipeline().parameters.dw_zone_table...
☐ dw_zone_table_pk_col	@pipeline().parameters.dw_zone_table...
☐ staging_zone_databas	@pipeline().parameters.staging_zone_...
☐ staging_zone_table_n:	@pipeline().parameters.staging_zone_t...
☐ staging_zone_table_pl	@pipeline().parameters.staging_zone_t...

All The Best 😊